

FTML practical session 16

12 juin 2026

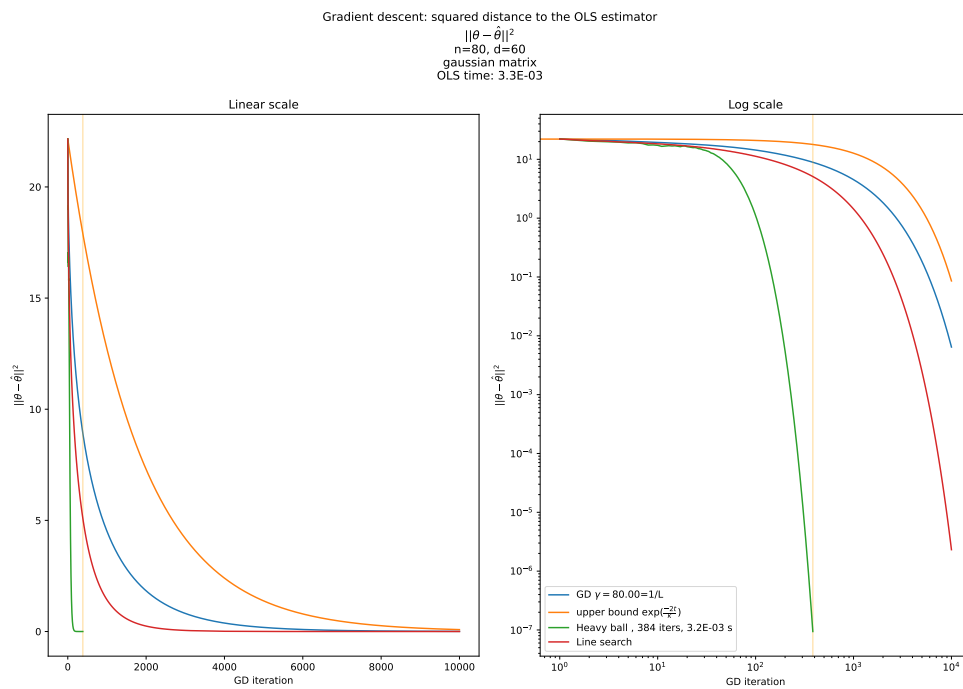


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1 THE HEAVY-BALL METHOD

1.1 Convergence rates of GD for convex functions

We consider the optimization of a convex function $f : \theta \rightarrow f(\theta)$ using a gradient descent (GD). In particular, we consider the **convergence speed** of GD. This speed can be expressed in several manners. For instance, as the distance between the iterate θ_t and a minimizer θ^* (of course assuming that this minimizer exists and is unique), as a function of the iteration number t . It is possible to show the following results for two-times differentiable convex functions :

- if H is invertible ($\mu > 0$), we have a convergence rate in $\exp(-\frac{2t}{\kappa})$.
- if H is not invertible ($\mu = 0$), we have a convergence rate in $\mathcal{O}(\frac{1}{t})$ (probably one of the exercises of the project).

These rates are speed **upper bounds**, meaning that the convergence is at least as fast as those. Also note that these rates of convergence require the use of specific values of the learning rate γ , like for instance $\frac{1}{L}$ (but other values might also be used, depending on the context). We also see that these rates depend on the **condition number** of the Hessian H . If we note μ the smallest eigenvalue of the Hessian H , and L the largest, and if this Hessian is for instance symmetric and definite positive, then

$$\kappa = \frac{L}{\mu} \quad (1)$$

However, the condition number might be defined also for general matrices, and even functions.

https://en.wikipedia.org/wiki/Condition_number

1.1.1 Large condition numbers

Hence, when κ is very large ($\gg 1$), the convergence to the optimum might be very slow. Note that matrices with large condition numbers are not rare in large-scale machine learning and scientific computing applications. If the smallest eigenvalue μ of a given matrix H is very small, or even 0 (which will happen as soon as H is not full rank), κ will be very large as soon as the largest eigenvalue L is not also very small.

1.1.2 Inertial methods

When κ is large, some methods still exist in order to speed the convergence of gradient descent, such as **Heavy-ball**. This method consists in adding a **momentum term** to the gradient update term, such as the iteration now writes

$$\theta_{t+1} = \theta_t - \gamma \nabla_{\theta} f(\theta_t) + \beta(\theta_t - \theta_{t-1}) \quad (2)$$

where β and γ are real constants that should be tuned. The update $\theta_{t+1} - \theta_t$ is then a combination of the gradient $\nabla_{\theta} f(\theta_t)$ and of the previous update $\theta_t - \theta_{t-1}$. The goal of this method is to balance the effect of oscillations in the gradient. The heavy-ball method is called an **inertial method**. When f is a general convex function (not necessary quadratic), some generalizations exist, such as **Nesterov acceleration**. Many of the most famous variations of SGD, like RMSProp and Adam, optionally include such a momentum term.

1.2 Simulation

Using `exercise_1_heavy_ball/main.py`, try to manually find γ and β values such that the convergence is faster with the Heavy-Ball method than with a standard GD.

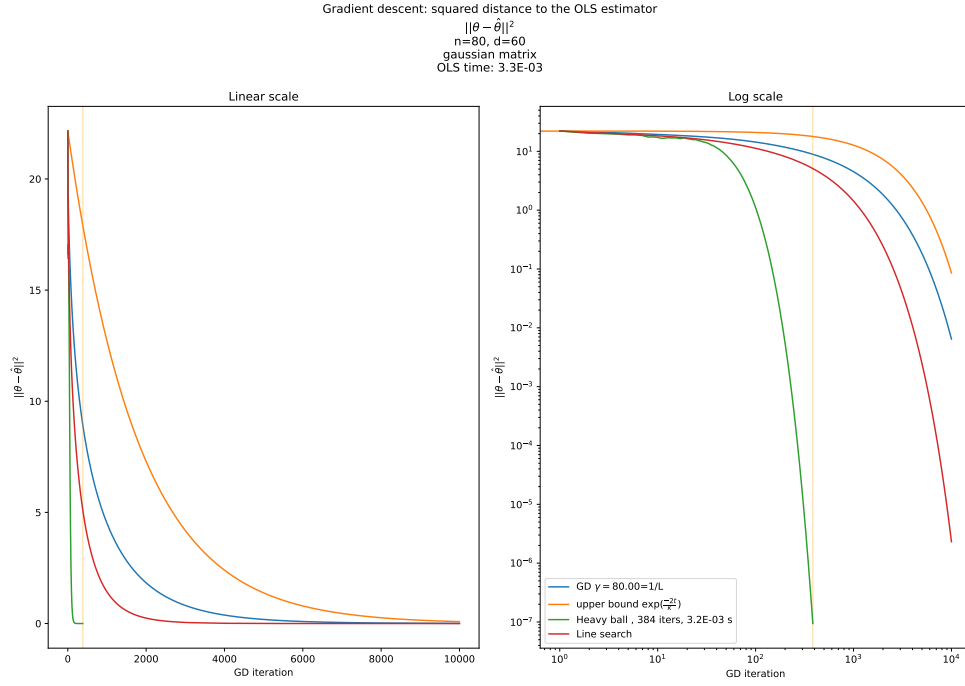


FIGURE 1 – Heavy ball vs GD, semilog scale

1.3 Analytical proof of γ and β values

Assuming $\mu > 0$, in a least squares problem, it is possible to show that the characteristic convergence time with the heavy-ball momentum term is $\sqrt{\kappa}$ instead of κ , **if β and γ are tuned well**. With the heavy-ball momentum term, we can change the convergence (upper bound) from $\mathcal{O}(\exp(-\frac{2t}{\kappa}))$ to $\mathcal{O}(\exp(-\frac{2t}{\sqrt{\kappa}}))$. If κ is large, which is the case we are interested in, this can be a significant improvement.

The update $\theta_{t+1} - \theta_t$ is then a combination of the gradient $\nabla_{\theta} f(\theta_t)$ and of the previous update $\theta_t - \theta_{t-1}$. This method might balance the effect of oscillations in the gradient. We will use these parameters :

$$\gamma = \frac{4}{(\sqrt{L} + \sqrt{\mu})^2} \quad (3)$$

and

$$\beta = \left(\frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}} \right)^2 \quad (4)$$

We keep the same notations as in the former practical sessions dedicated to the least squares problem. μ is the smallest eigenvalue of the Hessian H and L is the largest. Assuming $\mu > 0$ (strongly convex function), we will show that the characteristic convergence time with the heavy-ball momentum term is $\sqrt{\kappa}$ instead of κ .

Let λ be an eigenvalue of H and u_{λ} a eigenvector for this eigenvalue. We are interested in the evolution of $\langle \theta_t - \theta^*, u_{\lambda} \rangle$.

We note

$$a_t = \langle \theta_t - \theta^*, u_{\lambda} \rangle \quad (5)$$

Exercise 1 : Show that

$$a_{t+1} = (1 - \gamma\lambda + \beta)a_t - \beta a_{t-1} \quad (6)$$

Exercise 2: Compute the constant-recursive sequence a_t , and show that there exists a constant C_λ that depends on the initial conditions, such that

$$\forall t, a_t \leq (\sqrt{\beta})^t C_\lambda \quad (7)$$

https://en.wikipedia.org/wiki/Constant-recursive_sequence

If u_i is a basis of orthonormal vectors with eigenvalues λ_i , we have that

$$\begin{aligned} \|\theta_t - \theta^*\|^2 &= \sum_{i=1}^d (\langle \theta_t - \theta^*, u_i \rangle)^2 \\ &\leq \sum_{i=1}^d (\sqrt{\beta})^{2t} C_{\lambda_i} \\ &= (\sqrt{\beta})^{2t} D \end{aligned} \quad (8)$$

with

$$D = \sum_{i=1}^d C_{\lambda_i} \quad (9)$$

We can now remark that

$$\begin{aligned} \sqrt{\beta} &= \frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}} \\ &= \frac{1 - \sqrt{\frac{\mu}{L}}}{1 + \sqrt{\frac{\mu}{L}}} \\ &\leq 1 - \sqrt{\frac{\mu}{L}} \\ &= 1 - \frac{1}{\sqrt{\kappa}} \end{aligned} \quad (10)$$

Exercise 3: Conclude